

Some Harmonic Analysis

This is a direct extract from my algebra notes that I wanted to make a stand alone blog, see [2].

1 *Application to Harmonic Analysis

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters. Label this collection $\hat{G} = \text{Hom}_{\text{Ab}}(G, \mathbb{C}^*)$. This is a group with point-wise multiplication.

Proposition 1.1: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = (\text{Hom}_{\text{Ab}}(G, \mathbb{C}^*), \cdot)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \hat{\hat{G}}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \hat{\hat{G}}$ (think $\widehat{L^p} \cong L^q$)

Proof :

As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 | n_2 | \cdots | n_k$. Choose generators, $a_1, \dots, a_k$. Define the indicator characters on the generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

Clearly this is an isomorphism.

For the double-dual, take the natural map:

$$\Phi(g_i) = \text{ev}_{g_i}$$

where as usual $\text{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

This lets us understand the group-operation of G using point-wise function multiplication. Let us

now take a different approach. We can consider all “continuous” functions $f : G \rightarrow (\mathbb{C}, +)$. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. A-priori, this collection has no memory of the group operation of G . However, it is clearly in bijective correspondence to the $\mathbb{C}$ -vector space $\mathbb{C}[G]$ via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)_i$$

We want a $\mathbb{C}$ -algebra isomorphism between $\mathbb{C}[G]$ and $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. This preserves the addition of $\mathbb{C}[G]$ map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned} & \Phi \left(\sum_i a_i g_i + \sum_i b_i g_i \right) \\ &= \Phi \left(\sum_i (a_i + b_i) g_i \right) \\ &\mapsto (g_i \mapsto a_i + b_i) \\ &= (g_i \mapsto a_i) + (g_i \mapsto b_i) \\ &= \Phi \left(\sum_i a_i g_i \right) + \Phi \left(\sum_i b_i g_i \right) \end{aligned}$$

which should be intuitive since addition was defined formally. On the other hand, it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned} & \Phi \left(\left(\sum_{g \in G} a_i g_i \right) \left(\sum_{g \in G} b_i g_i \right) \right) \\ &= \Phi \left(\sum_{g \in G} \sum_{g_k g_\ell = g} (a_k b_\ell) g \right) \\ &\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell)) \\ &\neq (g_i \mapsto a_i b_i) \\ &\neq (g_i \mapsto a_i) \cdot (g_i \mapsto b_i) \\ &= \Phi \left(\sum_i a_i g_i \right) \cdot \Phi \left(\sum_i b_i g_i \right) \end{aligned}$$

And it is impossible to define a group-structure on G that will make this an equality unless $G = \{e\}$. This is because the points g_i act on each other. There is a natural multiplication function defined on $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ called *convolution*:

Definition 1.2: Convolution

Let $p, q \in \text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i) q(g_j) = \sum_{h \in G} p(h) q(h^{-1}g)$$

Lemma 1.3: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G], +, \cdot) \cong_{k\text{-}\mathbf{Alg}} (\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

Proof :

refer to the above calculations.

Now, as $G \cong \hat{G}$ and G forms an orthonormal basis for $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ via indicator functions, it is natural to ask whether $\hat{G}$ can form such a basis. First, G representing an orthonormal basis is equivalent to defining “evaluation” for $f : G \rightarrow \mathbb{C}$ using the usual dot product. To see this, consider:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Then for any $f : G \rightarrow \mathbb{C}$ there is a unique decomposition (or finite series):

$$f = \sum_{h \in G} f(h) \delta_h \quad \text{so that} \quad f(g) = \sum_{h \in G} f(h) \delta_h(g) = f(g) \cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product: by taking the unique inner product (up to isomorphism):

$$\langle f, g \rangle = \sum_{h \in G} f(h) \overline{g(h)}$$

Then $\{\delta_g\}_{g \in G}$ is an orthogonal basis and:

$$\langle f, \delta_{g_i} \rangle = \sum_{h \in G} f(h) \cdot \overline{\delta_{g_i}(h)} = f(g_i) \cdot 1 = f(g_i) = a_i$$

this “evaluation” behaviour of the inner product with the indicator is because the inner product is just the dot product, which in this case becomes a projection function

$$\langle f, \delta_{g_i} \rangle = f(g_i) \delta_{g_i}$$

This is just a special case of Riesz Representation Theorem that says that any linear functional is inner-product representable using these δ_{g_i} .

Now, the characters $\hat{G}$ are not *a-priori* orthogonal like the indicator functions given by G : one of the key results of harmonic analysis is that they do!

Proposition 1.4: Character Form Orthogonal Basis: Fourier Basis

The characters $\hat{G}$ form an orthogonal basis for $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

Proof :

If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i \rangle &= \sum_{g \in G} \chi_i(g) \overline{\chi_i(g)} \\ &= \sum_{g \in G} |\chi_i(g)|^2 \\ &= \sum_{g \in G} 1^2 \\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i \chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j \rangle = S = \sum_{g \in G} \chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h) \sum_{g \in G} \chi'(g) = \sum_{g \in G} \chi'(h)\chi'(g) = \sum_{g \in G} \chi'(h+g)$$

As $h + G = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g \in G} \chi'(g) = S \quad \text{so} \quad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j \rangle = 0$$

as we sought to show.

As the indicator functionals represent evaluation, the question arises what taking the inner product with the characters represent. This is most commonly thought of as extracting the “phase” information.

Example 1.5: Phase-Space Basis

Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0), (1,0), (0,1), (1,1)\}$. A character $\chi : G \rightarrow \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, we must have $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2 . Thus, the character’s values on the generators must be either 1 or -1 . This gives $2 \times 2 = 4$ possible characters, which we can index by pairs $(k_1, k_2) \in \{0,1\}^2$.

The four characters χ_{k_1, k_2} are defined by their values on the generators:

$$\chi_{k_1, k_2}(1,0) = (-1)^{k_1} \quad \text{and} \quad \chi_{k_1, k_2}(0,1) = (-1)^{k_2}$$

From this, the value for any element $(a,b) \in G$ is $\chi_{k_1, k_2}(a,b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new “basis”:

$\hat{G}$	(0,0)	(1,0)	(0,1)	(1,1)
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

We can verify the orthogonality relation from proposition 1.4 directly. Let us compute $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}
\langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g) \overline{\chi_{11}(g)} \\
&= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\
&= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\
&= 1 + 1 - 1 - 1 = 0
\end{aligned}$$

Using the above example, let us change perspective's on how to interpret this change of basis. The function $f : G \rightarrow \mathbb{C}$ can be thought of as taking the “points” $g \in G$ and mapping them (continuously) to the complex numbers. These points are evaluated by the use of a Riesz-representation of f using the indicator functions that form an orthonormal basis. But we just found another orthonormal basis for $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ with which we can take the inner product. We may thus think of the characters $\chi \in \hat{G}$ as points in their own right: these points are represent “phases” information rather than “location” information that the indicator functions represent. This motivates the following definition:

Definition 1.6: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$(\hat{}) : \text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \mapsto \text{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}) \quad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g) \overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Lebesgue measurable functions. It wasn't until later where mathematicians like Folland and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Now, certainly as vector spaces $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \cong_{\mathbb{C}} \text{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$ mapping:

$$(g \mapsto \langle f, \delta_g \rangle) \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

But does this map preserve the inner product, and more so can the convolution product of $\text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ be preserved given the appropriate choice of product on $\text{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$? This answer is yes and can be thought as the corner-stone result of harmonic analysis on discrete abelian groups:

Theorem 1.7: Fourier Reciprocity

The map given in definition 1.6 is a unitary $\mathbb{C}$ -algebra isomorphism,

$$(\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *) \cong (\text{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$$

with $\cdot$ in $(\text{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$ being point-wise multiplication.

Recall that unitary means $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$. In a conventional real analysis course:

- the $\mathbb{C}$ -algebra homomorphism is called the *Convolution Theorem*,
- the unitary property is called the *Plancherel's Theorem*,
- invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

Proof :

Some of these are essentially already proven, but for the completeness of this theorem all will be laid out.

1. *Linearity*: Let $f, h \in \text{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ and $a, b \in \mathbb{C}$.

$$\begin{aligned} \widehat{af + bh}(\chi) &= \langle af + bh, \chi \rangle && \text{by definition} \\ &= \sum_{g \in G} (af + bh)(g) \overline{\chi(g)} \\ &= a \sum_{g \in G} f(g) \overline{\chi(g)} + b \sum_{g \in G} h(g) \overline{\chi(g)} && \text{by linearity of sums} \\ &= a \langle f, \chi \rangle + b \langle h, \chi \rangle \\ &= a \hat{f}(\chi) + b \hat{h}(\chi) \end{aligned}$$

Since this holds for all $\chi \in \hat{G}$, the transform is a linear map.

2. *$\mathbb{C}$ -algebra Homomorphism (The Convolution Theorem)*: We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned} \widehat{f * h}(\chi) &= \langle f * h, \chi \rangle \\ &= \sum_{k \in G} (f * h)(k) \overline{\chi(k)} && \text{by definition of inner product} \\ &= \sum_{k \in G} \left(\sum_{g \in G} f(g) h(g^{-1}k) \right) \overline{\chi(k)} && \text{by definition of convolution} \\ &= \sum_{g \in G} f(g) \sum_{k \in G} h(g^{-1}k) \overline{\chi(k)} && \text{swapping order of summation} \end{aligned}$$

Let $\ell = g^{-1}k$, so $k = g\ell$. As k runs over G , so does ℓ . The inner sum becomes:

$$\sum_{\ell \in G} h(\ell) \overline{\chi(g\ell)} = \sum_{\ell \in G} h(\ell) \overline{\chi(g)} \overline{\chi(\ell)} = \overline{\chi(g)} \sum_{\ell \in G} h(\ell) \overline{\chi(\ell)} = \overline{\chi(g)} \hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}\widehat{f * h}(\chi) &= \sum_{g \in G} f(g) \left(\overline{\chi(g)} \hat{h}(\chi) \right) \\ &= \hat{h}(\chi) \sum_{g \in G} f(g) \overline{\chi(g)} \\ &= \hat{h}(\chi) \hat{f}(\chi)\end{aligned}$$

Thus, $\widehat{f * h} = \hat{f} \cdot \hat{h}$, proving the map is an algebra homomorphism.

3. *Isomorphism (The Fourier Inversion Theorem)*: We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g)$.

$$\begin{aligned}\frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g) &= \frac{1}{|G|} \sum_{\chi \in \hat{G}} \left(\sum_{h \in G} f(h) \overline{\chi(h)} \right) \chi(g) && \text{by definition of } \hat{f} \\ &= \frac{1}{|G|} \sum_{h \in G} f(h) \sum_{\chi \in \hat{G}} \chi(g) \overline{\chi(h)} && \text{swapping order of summation} \\ &= \frac{1}{|G|} \sum_{h \in G} f(h) \sum_{\chi \in \hat{G}} \chi(gh^{-1})\end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if $x = e$ and 0 otherwise. Thus $\sum_{\chi \in \hat{G}} \chi(gh^{-1}) = |G| \delta_{g,h}$.

$$\frac{1}{|G|} \sum_{h \in G} f(h) (|G| \delta_{g,h}) = \sum_{h \in G} f(h) \delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. *Unitary Property (Plancherel's Theorem)*: Note that here is where it is important for the inner product to be normalized. Let us calculate the inner product for the given

definition.

$$\begin{aligned}
\langle \hat{f}, \hat{h} \rangle_{\hat{G}} &= \sum_{\chi \in \hat{G}} \hat{f}(\chi) \overline{\hat{h}(\chi)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g) \overline{\chi(g)} \right) \overline{\left(\sum_{k \in G} h(k) \overline{\chi(k)} \right)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g) \overline{\chi(g)} \right) \left(\sum_{k \in G} \overline{h(k)} \chi(k) \right) \\
&= \sum_{g \in G} \sum_{k \in G} f(g) \overline{h(k)} \sum_{\chi \in \hat{G}} \chi(k) \overline{\chi(g)} && \text{re-arranging terms} \\
&= \sum_{g \in G} \sum_{k \in G} f(g) \overline{h(k)} (|G| \delta_{g,k}) && \text{by Second Orthogonality} \\
&= |G| \sum_{g \in G} f(g) \overline{h(g)} \\
&= |G| \langle f, h \rangle_G
\end{aligned}$$

This proves *Plancherel's Theorem*. To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}} \hat{f}$, then it follows immediately that $\langle \mathcal{U}(f), \mathcal{U}(h) \rangle = \langle f, h \rangle$, completing the proof.

Example 1.8: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ from example 1.5. Let us now compute the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard “position” basis. We find its coordinates in the character basis by computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned}
\hat{f}(\chi) &= \langle \delta_{(1,0)}, \chi \rangle = \sum_{g \in G} \delta_{(1,0)}(g) \overline{\chi(g)} \\
&= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0)
\end{aligned}$$

Using the character table, we can compute this value for each χ :

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the “position” basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 1.7: $f(g) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g)$ (here

$|G| = 4$). Let's check the value at $g = (1, 0)$:

$$\begin{aligned} f(1, 0) &= \frac{1}{4} \left(\hat{f}(\chi_{00})\chi_{00}(1, 0) + \hat{f}(\chi_{10})\chi_{10}(1, 0) + \hat{f}(\chi_{01})\chi_{01}(1, 0) + \hat{f}(\chi_{11})\chi_{11}(1, 0) \right) \\ &= \frac{1}{4} ((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)) \\ &= \frac{1}{4} (1 + 1 + 1 + 1) = 1 \end{aligned}$$

Now let's check the value at $g = (0, 1)$, where we expect 0:

$$\begin{aligned} f(0, 1) &= \frac{1}{4} \left(\hat{f}(\chi_{00})\chi_{00}(0, 1) + \hat{f}(\chi_{10})\chi_{10}(0, 1) + \hat{f}(\chi_{01})\chi_{01}(0, 1) + \hat{f}(\chi_{11})\chi_{11}(0, 1) \right) \\ &= \frac{1}{4} ((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)) \\ &= \frac{1}{4} (1 - 1 - 1 + 1) = 0 \end{aligned}$$

Let us now connect this to Representation Theory. So far, we have established that for a finite abelian group G , the Fourier transform is a unitary $\mathbb{C}$ -algebra isomorphism between the algebra of functions on G with convolution, $(\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of a Artin-Wedderburn decomposition.

First, take the regular representation of G , that is left-multiplication (convolution) $\rho : G \rightarrow \text{GL}(\mathbb{C}[G])$ given by:

$$\rho(g) : f \mapsto g * f$$

Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the Pontryagin dual: $\text{Irr}(G) = \hat{G}$ and so we get:

$$\mathbb{C}[G] \cong_{\mathbb{C}} \bigoplus_{\chi \in \hat{G}} V_{\chi}$$

where each V_{χ} is a 1-dimensional subspace corresponding to the character χ .

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi)\chi$ is the projection of f onto the 1-dimensional irreducible subspace V_{χ} . Then the Fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ . Note that we can define projection functions $p_{\chi} = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)}g$ and in $\mathbb{C}[G]$ these project onto the irreducible subspace V_{χ} . Convolution with p_{χ} annihilates all other “frequency components”, leaving the part of f that lies in V_{χ} . Hence, the Fourier basis *the* unique basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the concept of a dual group $\hat{G}$ and a simple transform onto it ceases to be useful (it is now a set with no natural group structure). However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

1.1 Generalization to Locally Compact Abelian Groups

What happens when we wish to analyze functions on abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting requires us to sum on an arbitrary number of elements which introduces the usual problem of undefinedness. To tame the infinite behaviour, notice that in the finite case we can think of the delta functions as *measures* so that any f is a sum of weighted singular measures. Thus, we aim to introduce a natural measure on abelian groups preserving the group structure. The essential new ingredient is the *Haar measure*. For any LCA group G , there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 1.9: Haar Measure

A *left-Haar measure* on an Locally compact (not necessarily abelian) group G is a measure μ on the Borel σ -algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. A similar definition works for the *right-Haar measure*. For abelian groups, it is also right-translation-invariant, and it is simply called the *Haar measure*.

See [1] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 1.10: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx .
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

Note that in the infinite case, we can no longer represent points as indicator functions. Instead the *direct-delta distributions* would take this role; see [1] for more detail.

With the Haar measure, we can redefine our main objects of study by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$* : The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group $\widehat{G}$* : It was proven that the correct codomain should be S^1 (see [ref:HERE](#) for the details, Poincare duality paper). The dual group is the set of all *continuous* group homomorphisms into S^1 .

$$\widehat{G} = \text{Hom}_{\text{Top}}(G, S^1)$$

The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, it is rarely true that $G \cong \widehat{\widehat{G}}$. For example:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \quad \widehat{\mathbb{Z}} \cong S^1, \quad \widehat{S^1} \cong \mathbb{Z}$$

This highlights why it is essential to treat $\widehat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform:* The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\widehat{G}$:

$$\hat{f}(\chi) = \int_G f(g) \overline{\chi(g)} d\mu(g)$$

Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h \rangle = \int_G f(g) \overline{h(g)} d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 1.11: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ . Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ such that the Fourier transform, appropriately normalized, is a unitary $\mathbb{C}$ -algebra isomorphism. The main results are:

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:

$$f(g) = \int_{\widehat{G}} \hat{f}(\chi) \chi(g) d\nu(\chi)$$

2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \rightarrow L^2(\widehat{G}, \nu)$. This means it preserves the inner product:

$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$

3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:

$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

Proof :

see [cite:HERE](#) (it's not in my analysis book, and I didn't type it up in Pontryagin notes)

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	LCA Case
Group G	LCA Group G
Order $ G $	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\text{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\hat{G} = \text{Hom}(G, \mathbb{C}^\times)$	Dual Group $\hat{G} = \text{Hom}_{\mathbf{Top}}(G, S^1)$
Inner Product $\sum f(g)\overline{h(g)}$	Inner Product $\int f(g)\overline{h(g)}d\mu(g)$
Fourier Transform $\sum f(g)\chi(g)$	Fourier Transform $\int f(g)\chi(g)d\mu(g)$

Further interesting facts can be found here:

<https://mathoverflow.net/questions/37021/is-fourier-analysis-a-special-case-of-representation-theory>

1.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n , the corresponding algebra is the polynomial ring $k[x_1, \dots, x_n]$. The points of the space correspond to the maximal ideals of this algebra. Specifically, for any point $p = (a_1, \dots, a_n) \in k^n$, the evaluation map:

$$\text{ev}_p : k[x_1, \dots, x_n] \rightarrow k \quad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \dots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \dots, x_n]/m_p \cong k$ recovers the value of the function at that point.

We have established a similar setup for a finite abelian group G , with the convolution algebra $(\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the “points” of our space from the maximal ideals of this algebra? Let’s first attempt a direct analogy. We might assume the points are the elements of the group G itself. For a point $g_i \in G$, we can define the point-evaluation map, which is a linear map:

$$\text{ev}_{g_i} : \text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \rightarrow \mathbb{C} \quad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i . However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\text{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k)f(k^{-1}g_i)$$

Just because $f(g_i) = 0$ does not imply that the sum on the right-hand side is zero. The value of f at other points, $f(k^{-1}g_i)$, can be non-zero. Therefore, the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct “geometric points” for the convolution algebra are not the elements of G , but the characters of the dual group $\hat{G}$.

Proposition 1.12: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \hat{G}$.

Proof :

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\text{ev}}_\chi : \text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \rightarrow \mathbb{C} \quad f \mapsto \widehat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ . By 1.7 $\widehat{\text{ev}}_\chi$ is a $\mathbb{C}$ -algebra homomorphism. Since $\widehat{\text{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ , is a maximal ideal:

$$I_\chi = \ker(\widehat{\text{ev}}_\chi) = \{f \in \text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \mid \widehat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}) / I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, After applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

For the infinite case, the maximal ideals of the Banach algebra $L^1(G, \mu)$ arise in this same way from homomorphisms into $\mathbb{C}$ (see the *Gelfand-Mazur Theorem*).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this “true” underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	Classical Algebraic Geometry
Algebra	Polynomial Ring $(k[x_1, \dots, x_n], \cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1, \dots, a_n) \in k^n$
Evaluation Map	$\text{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\text{ev}_p) = (x_i - a_i)$
Concepts	Harmonic Analysis on G
Algebra	Group Algebra $(\mathbb{C}[G], *)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi \in \widehat{G}$
Evaluation Map	Fourier "Evaluation" $\widehat{\text{ev}}_\chi : f \mapsto \widehat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\text{ev}}_\chi) = I_\chi$

References

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- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.